MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, LOW POWER SCHOTTKY, TTL, MONOSTABLE MULTIVIBRATORS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, low power Schottky, monostable multivibrator microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Dual monostable multivibrator, retriggerable, with clear
02	Dual monostable multivibrator, Schmitt trigger inputs, with clear
03	Single monostable multivibrator, retriggerable, with clear

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
B	F-3 (14-lead, 3/16" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line
D	F-2 (14-lead, 1/4" x 3/8"), flat package
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line
F	F-5 (16-lead, 1/4" x 3/8"), flat package
2	C-2 (20 terminal, .350" x .350") square chip carrier package

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to 7.0 V dc
Input voltage range	-1.5 V dc to -18 mA dc to 5.5 V dc
Storage temperature	-65°C to 150°C
Maximum power dissipation per multivibrator, (P_D) 1/	
Device type 01	110 mW dc
Device type 02	149 mW dc
Device type 03	61 mW dc
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction to case (θ_{JC}):	
Cases A, B, D, F	70°C/W
Cases C, E	50°C/W
Case 2	60°C/W
Junction temperature (T_J)	175°C

1.4 Recommended operating conditions.

Supply voltage	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH})	2.0 V dc
Maximum low level input voltage (V_{IL})	0.7 V dc
Case operating temperature range (T_C)	-55°C to 125°C
Minimum pulse width	40 ns
Input pulse rise/fall time, device type 02	
Schmitt, B input	1 V/s minimum
Logic, A input	1 V/µs minimum
Clear-inactive-state setup time	
Device type 02	15 ns minimum
External timing resistance, R_{ext}	
Device type 01, 03	5 kΩ minimum, 180 kΩ maximum
Device type 02	1.4 kΩ minimum, 70 kΩ maximum
External timing capacitance, C_{ext}	
Device type 01, 03	No restriction
Device type 02	1,000 µF maximum
Output duty cycle, device type 02	
R_T = 2 kΩ	50% duty cycle maximum
T_T = 70 kΩ	90% duty cycle maximum
Wiring capacitance, R_{ext}/C_{ext} terminal	
Device type 01, 03 (referenced to GND)	50 pF maximum

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IL} = 0.7 V, V_{IH} = 2.0 V, I_{OH} = -400 µA	01, 02, 03	2.5	---	Volts
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IL} = 0.7 V V_{IH} = 2.0 V, I_{OL} = 4 mA		---	0.4	
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA T_C = 25°C		---	-1.5	
Low-level input current	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01, 03	-160	-400	µA
Low-level input current at clear input	I_{IL2}		02	-30	-680	
Low-level input current at A_{IN}			02	-30	-340	
Low-level input current at B_{IN}			02	-30	-580	
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	01, 02, 03	---	20	µA
High-level input current	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		---	100	
Short circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{IH} = 5.5 V, V_{IL} = GND		-15	-130	mA
Supply current (quiescent)	I_{CC1}	V_{CC} = 5.5 V	02	---	11	mA
Supply current (quiescent or triggered)	I_{CC2}		01 03	--- ---	20 11	
Supply current (triggered)	I_{CC3}		02	---	27	
Propagation delay time, low-to-high level from input A	t_{PLH1}	V_{CC} = 5.0 V C_L = 50 pF ±10%	01, 03 02	5	57 113	ns
Propagation delay time, low-to-high level from input B	t_{PLH2}	R_L = 2 kΩ ±5% C_{ext} = 1/	01, 03 02		74 90	
Propagation delay time, low-to-high level from clear	t_{PLH3}	R_{ext} = 1/	01, 03 02		75 105	
Propagation delay time, high-to-low level from input A	t_{PHL1}		01, 03 02		75 128	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
Propagation delay time, high-to-low level from input B	t_{PHL2}	V_{CC} = 5.0 V C_L = 50 pF ±10%	01, 03	5	92	ns
			02		105	
Propagation delay time, high-to-low level from clear	t_{PHL3}	R_L = 2 kΩ ±5% C_{ext} = 1/ R_{ext} = 1/	01, 03		48	
			02		90	
Minimum pulse width of Q output	$t_{p(MIN)}$		01, 03	---	308	
Width of Q output pulse	t_{p1}	V_{CC} = 5.0 V	02	20	91	ns
	t_{p2}	C_L = 50 pF ±10%	02	70	195	ns
	t_{p3}	R_L = 2 kΩ ±5%	02	600	850	ns
	t_{p4}	C_{ext} = 2/ ±10%	01, 03	3.0	6.25	µs
	t_{p5}	R_{ext} = 2/ ±10%	02	5.5	8.5	ms

1/ For propagation delay tests, see table III for C_{ext} and R_{ext} values.

2/ $t_{p(MIN)}$ test, C_{ext} = open and R_{ext} = 5 kΩ.

t_{p1} test, C_{ext} = open and R_{ext} = 2 kΩ.

t_{p2} test, C_{ext} = 80 pF and R_{ext} = 2 kΩ.

t_{p3} test, C_{ext} = 100 pF and R_{ext} = 10 kΩ.

t_{p4} test, C_{ext} = 1,000 pF and R_{ext} = 10 kΩ.

t_{p5} test, C_{ext} = 1 µF and R_{ext} = 10 kΩ.

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.2.2 Truth table and functional description. The truth table and functional description shall be as specified on figure 2.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturers' schematics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 10 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*, 2, 3, 7, 9, 10, 11	1*, 2, 3, 7, 9
Group A test requirements (method 5005)	1, 2, 3, 7, 8, 9, 10, 11	1, 2, 3, 7, 8, 9
Group C end-point electrical parameters (method 5005)	N/A	1, 2, 3
Additional electrical subgroups for group C periodic inspections	N/A	10, 11
Group D end-point electrical parameters (method 5005)	1, 2, 3	1, 2, 3

*PDA applies to subgroup 1 (see 4.2c).

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4. PRODUCT ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D using the circuit shown on figure 3, or equivalent.

(2) T_A = 125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration - 1,000 hours, except as permitted by appendix B of MIL-M-38510, and method 1005 of MIL-STD-883.

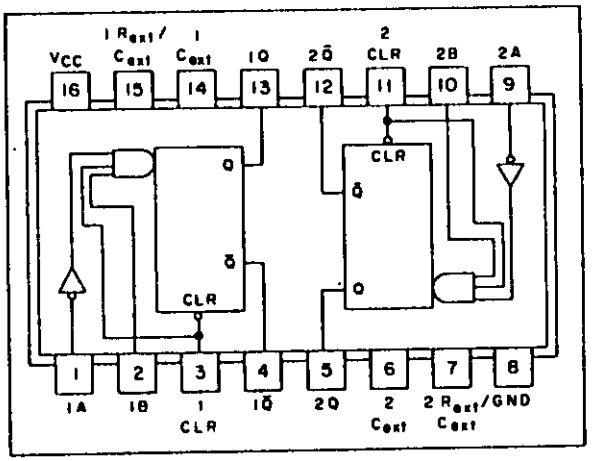
c. Subgroups 3 and 4 shall be added to group C inspection parameters for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

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Device type 01

Cases E and F



Case 2

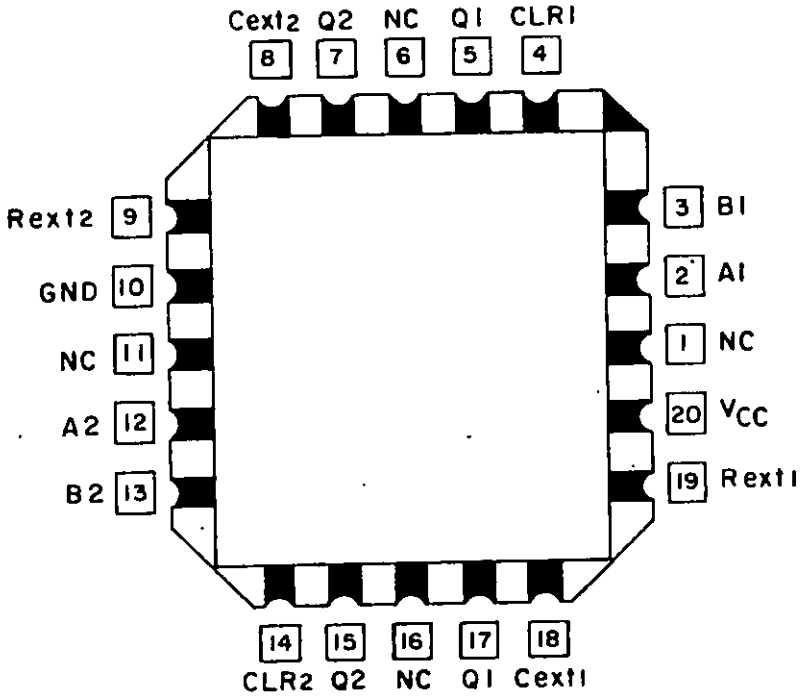
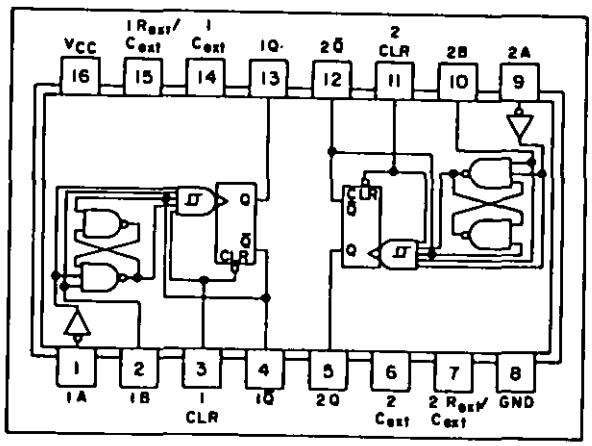


FIGURE 1. Logic diagrams and terminal connections (top view).

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Device type 02

Cases E and F



Case 2

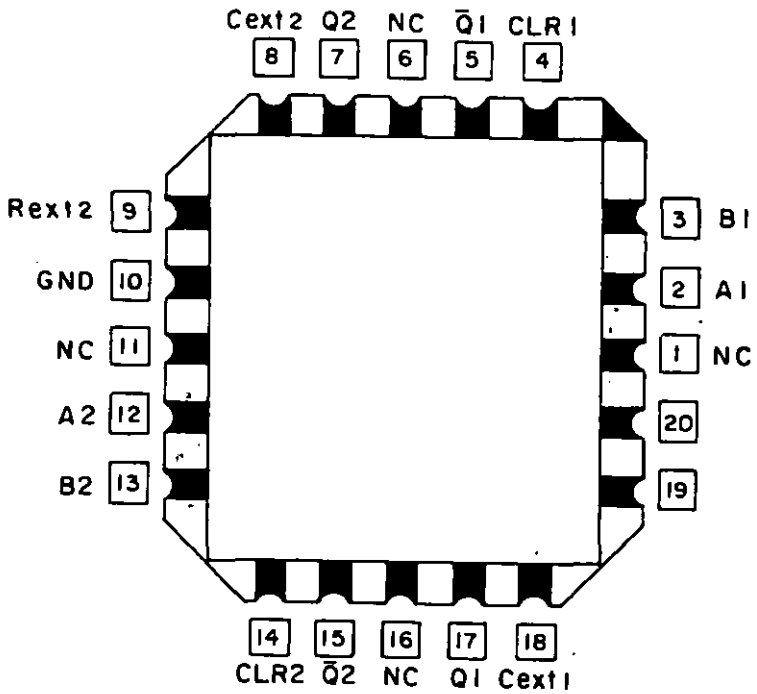
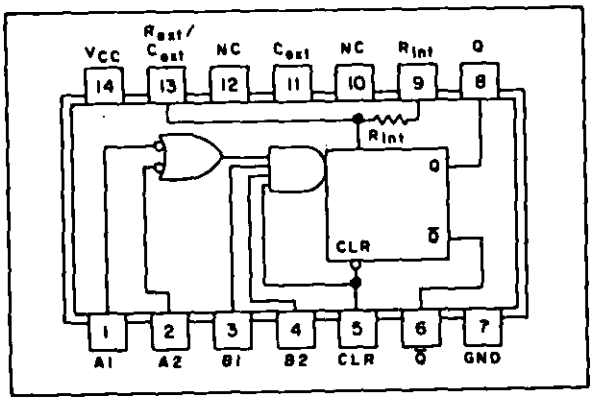


FIGURE 1. Logic diagrams and terminal connections (top view) - Continued.

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Device type 03

Cases A, B, C and D



Case 2

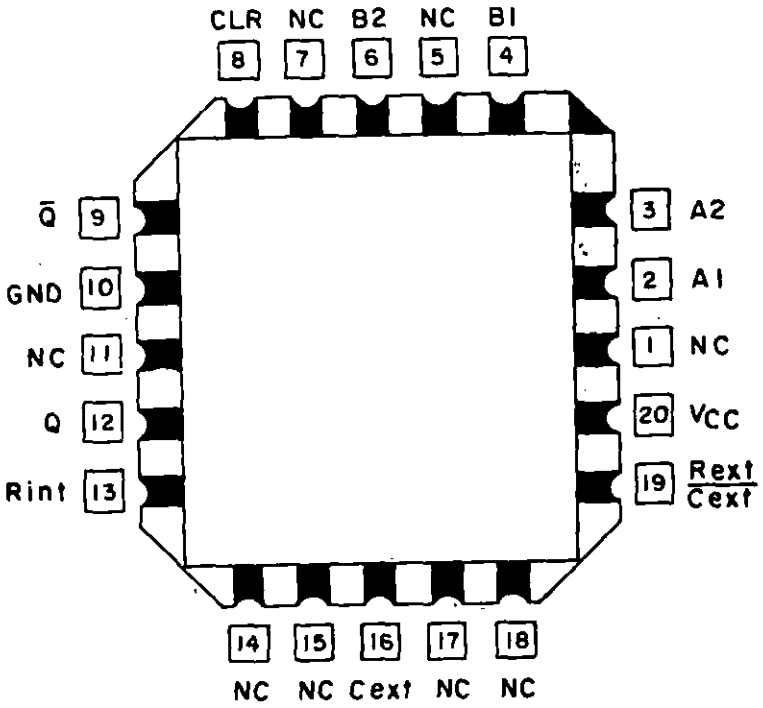


FIGURE 1. Logic diagrams and terminal connections (top view) - Continued.

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Device type 01

INPUTS			OUTPUTS	
CLEAR	A	B	Q	$\overline{Q}$
L	X	X	L	H
X	H	X	L	H
X	X	L	L	H
H	L	↑	⎍	⎎
H	↓	H	⎍	⎎
↑	L	H	⎍	⎎

Device type 02

INPUTS			OUTPUTS	
CLEAR	A	B	Q	$\overline{Q}$
L	X	X	L	H
X	H	X	L	H
X	X	L	L	H
H	L	↑	⎍	⎎
H	↓	H	⎍	⎎
↑	L	H	⎍	⎎

Device type 03

INPUTS					OUTPUTS	
CLEAR	A1	A2	B1	B2	Q	$\overline{Q}$
L	X	X	X	X	L	H
X	H	H	X	X	L	H
X	X	X	L	X	L	H
X	X	X	X	L	L	H
H	L	X	↑	H	⎍	⎎
H	L	X	H	↑	⎍	⎎
H	X	L	↑	H	⎍	⎎
H	X	L	H	↑	⎍	⎎
H	H	↓	H	H	⎍	⎎
H	↓	↓	H	H	⎍	⎎
H	↓	H	H	H	⎍	⎎
↑	L	X	H	H	⎍	⎎
↑	X	L	H	H	⎍	⎎

NOTES:
1. H = high level (steady state), L = low level (steady state), ↑ = transition from low to high level, ↓ = transition from high to low level, ⎍ = one high level pulse, ⎎ = one low level pulse, X = irrelevant (any input, including transitions).
2. To use the internal timing resistor of device 03 connect R_{int} to V_{CC}.
3. An external timing capacitor may be connected between C_{ext} and R_{ext}/C_{ext} (positive).
4. For accurate repeatable pulse widths, connect an external resistor between R_{ext}/C_{ext} and V_{CC} with R_{int} open circuited.
5. To obtain variable pulse widths, connect external variable resistance between R_{int} or R_{ext}/C_{ext} and V_{CC}.

FIGURE 2. Truth table and functional description.

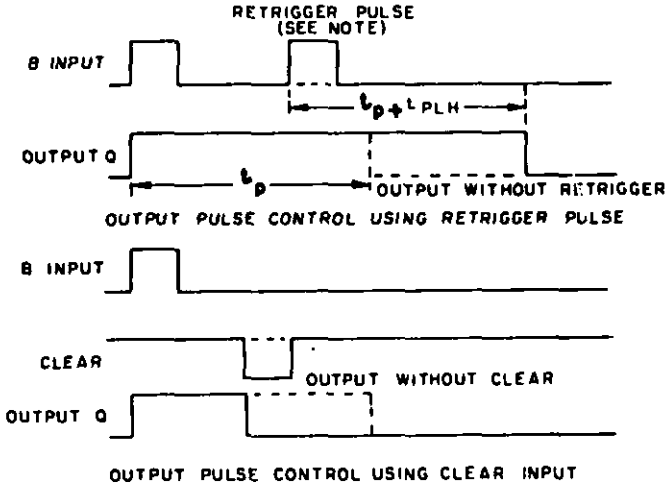
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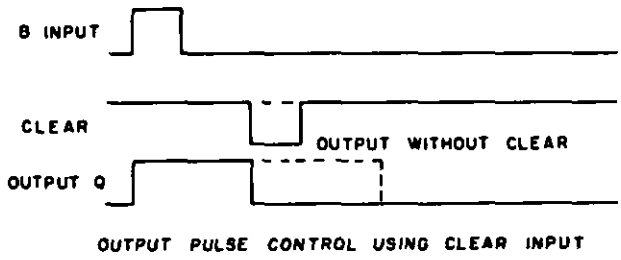
Device types 01 and 03

This multivibrator features d-c triggering from gated low-level-active (A inputs), and high-level-active (B inputs). Output pulse width is a function of external capacitor and resistor values. Retriggering of input before output terminates, extends output pulse width. Overriding clear feature permits termination of output pulse width at a predetermined time independent of R and C timing components.

Device type 02

This multivibrator features a negative-transition-triggered input and a positive-transition-triggered input, either of which can be used as an inhibit input. Pulse triggering occurs at a particular voltage level, not directly related to transition time of input pulse. Once fired, the outputs are independent of further transitions of A and B inputs, and are a function of the timing components. Output pulses can be terminated by the overriding clear, independent of R and C timing components.





NOTE: Retrigger pulse must not start before 0.22 C_{ext} (in picofarads) nanoseconds after previous trigger pulse.

FIGURE 2. Truth table and functional description - Continued.

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Device types 01 and 03 (Continued)

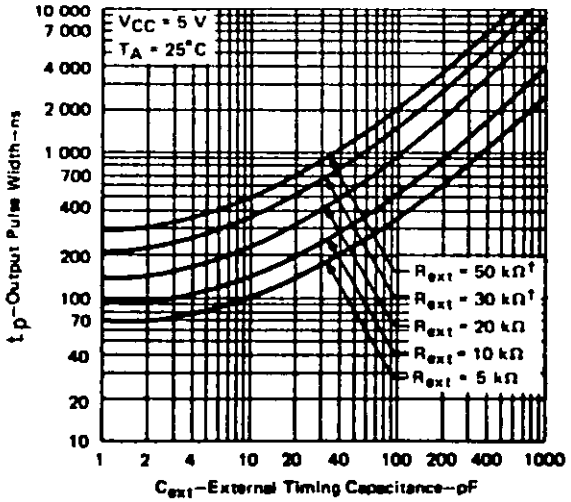
The output pulse width (t_p) is a function of the external capacitor and resistor values.

For output pulse widths when C_{ext} > 1,000 pF, t_p is defined as:

$t_p \approx 0.4\ R_T\ C_{ext}$
Where R_T is in kΩ,
C_{ext} is in pF,
and t_p is in ns.

For output pulse widths when C_{ext} < 1,000 pF, t_p is defined as:

TYPICAL OUTPUT PULSE WIDTH
vs
EXTERNAL TIMING CAPACITANCE



† These values of resistance exceed the maximum recommended for use over the full temperature range of device type 01.

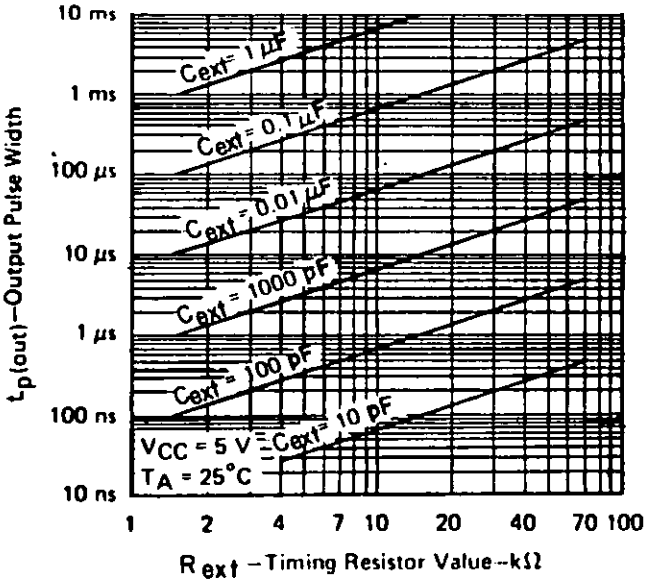
Device type 02 (Continued)

The output pulse width (t_p) is a function of the external capacitor and resistor values.

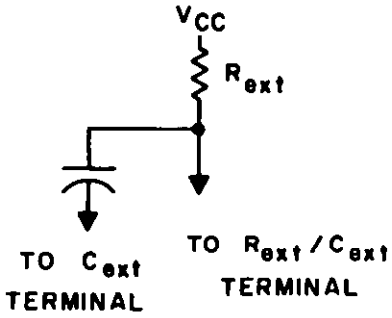
The output pulse width is defined as:

$t_p \approx 0.7\ R_T\ C_{ext}$
Where R_T is in kΩ,
C_{ext} is in pF,
and t_p is in ns.

OUTPUT PULSE WIDTH
vs
TIMING RESISTOR VALUE



Device types 01, 02 and 03

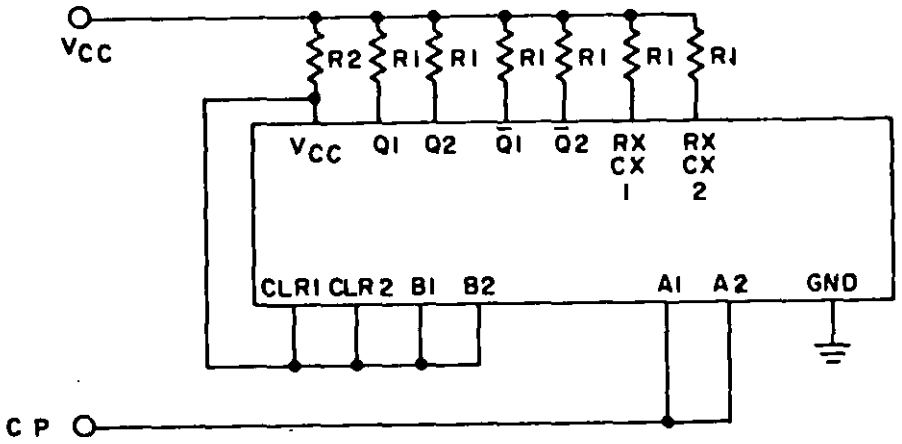


TIMING COMPONENT CONNECTIONS

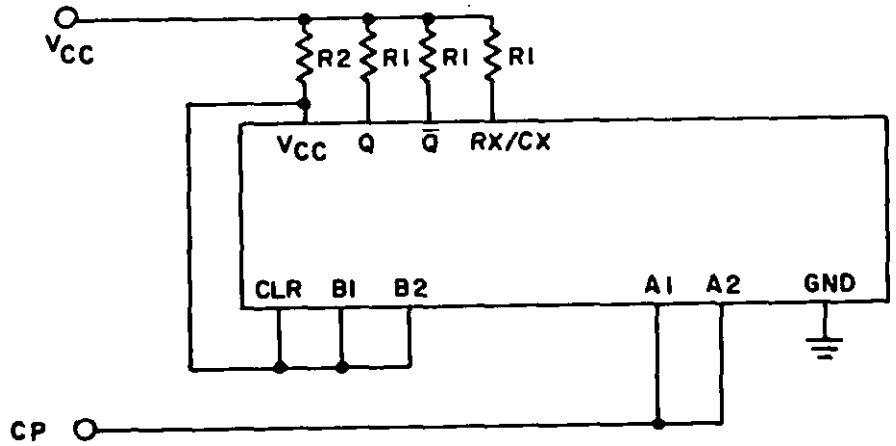
FIGURE 2. Truth table and functional description - Continued.

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Device types 01 and 02



Device type 03



NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 1 kΩ ±5%; R2 = 27Ω ±5%.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

FIGURE 3. Burn-in and life test circuit.

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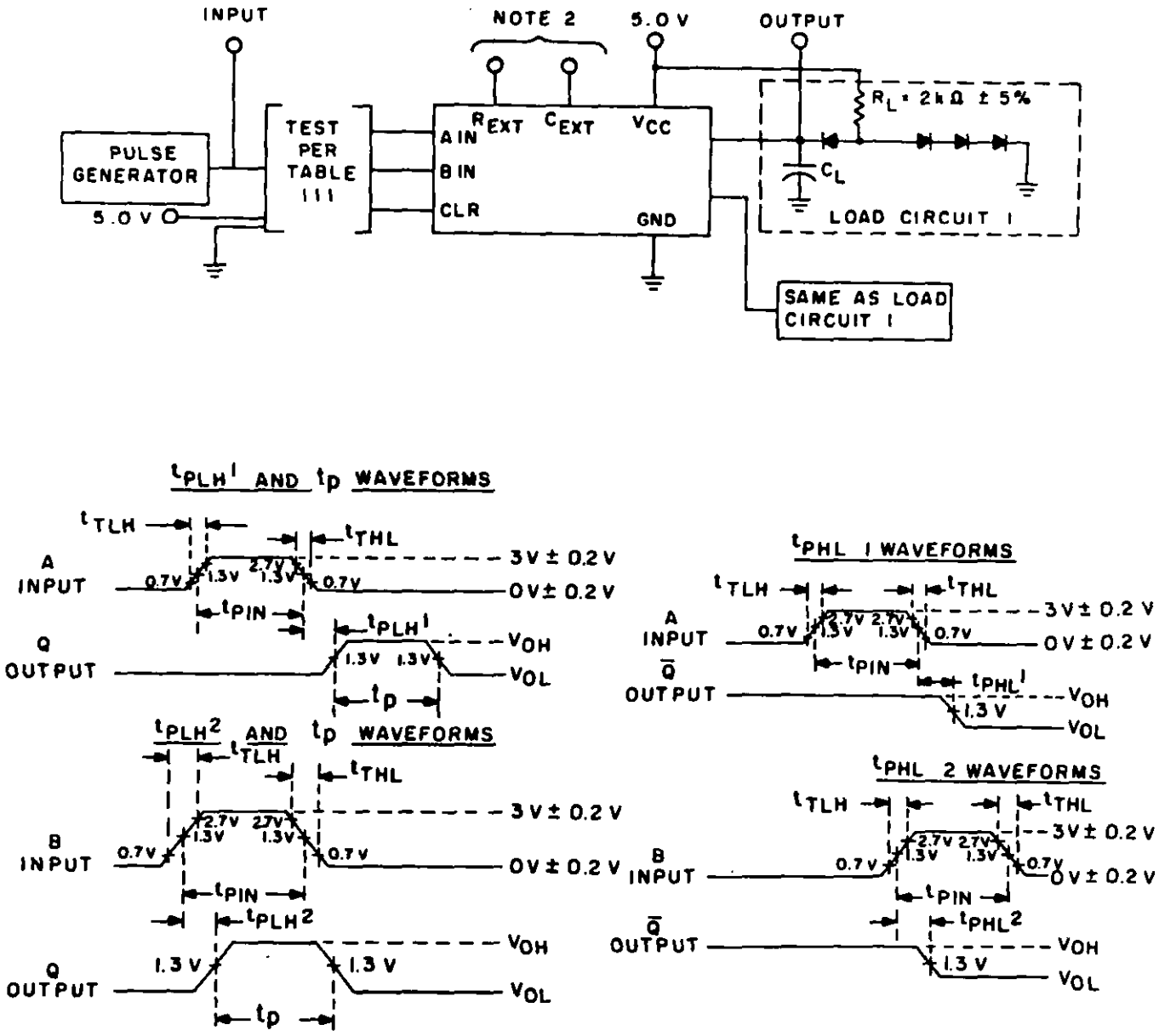
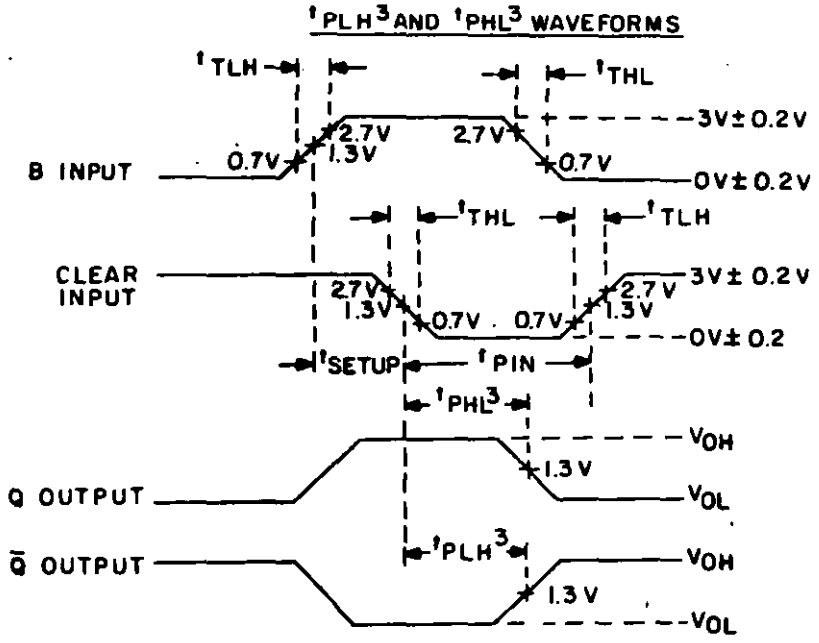


FIGURE 4. Switching test circuit and waveforms for device types 01 and 03.

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NOTES:
1. Pulse generator has the following characteristics: PRR ≤ 1.0 MHz, t_{PIN} ≥ 40 ns, t_{THL} ≤ 6 ns, t_{TLH} ≤ 15 ns and Z_{OUT} = 50Ω.
2. See table III notes for R_{ext}, C_{ext} values.
3. C_L = 50 pF ±10% including scope probe, wiring and stray capacitance, without package in test fixture.
4. All diodes are 1N3064 or 1N916 or equivalent.
5. Load circuit on a given output is only required where the specified test in table III indicates "OUT" on that output.
6. t_{SETUP} (max) shall be ≤ 50% of the typical output pulse width for the actual C_{ext} used (see figure 2).

FIGURE 4. Switching test circuit and waveforms for device types 01 and 03 - Continued.

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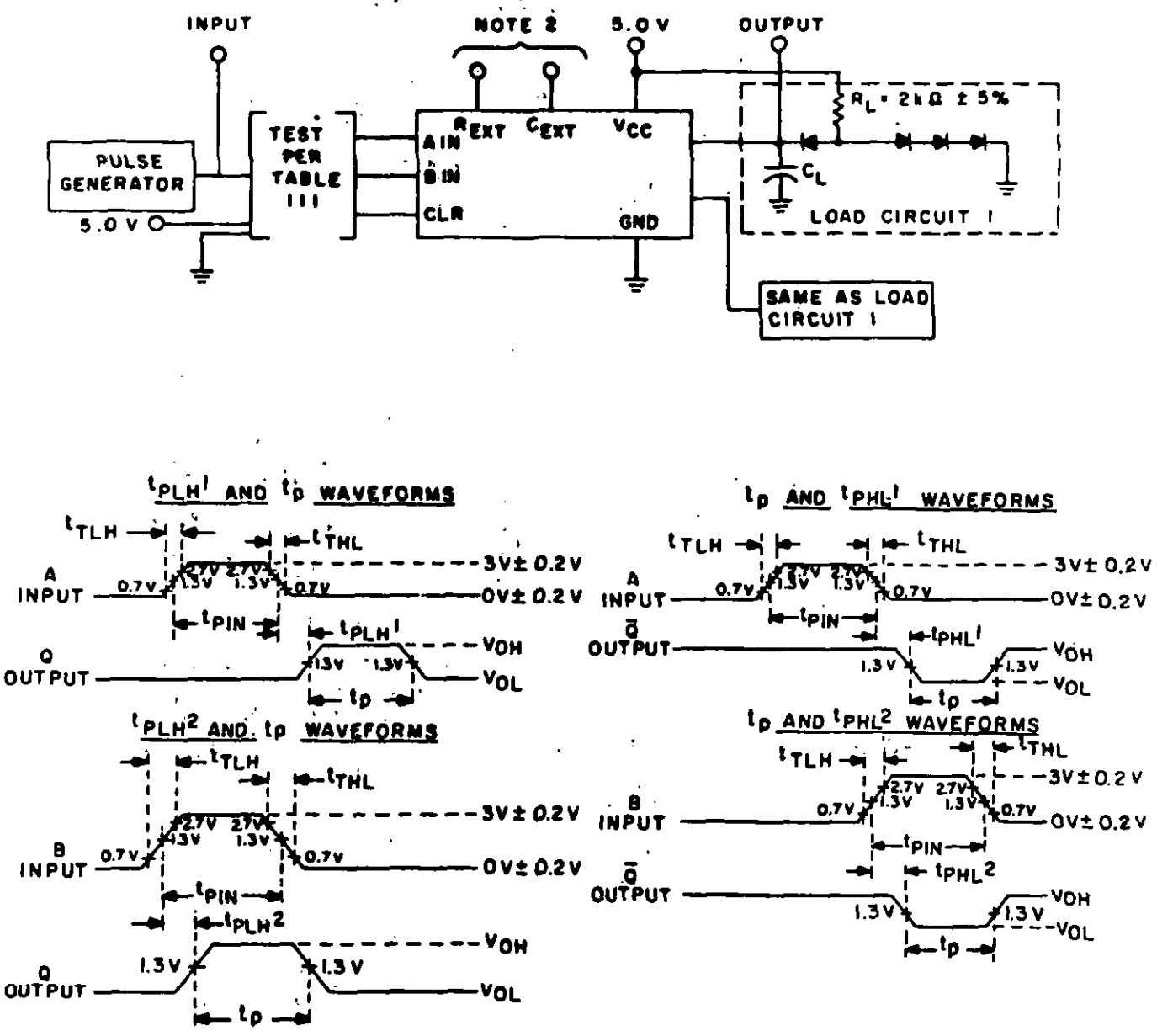
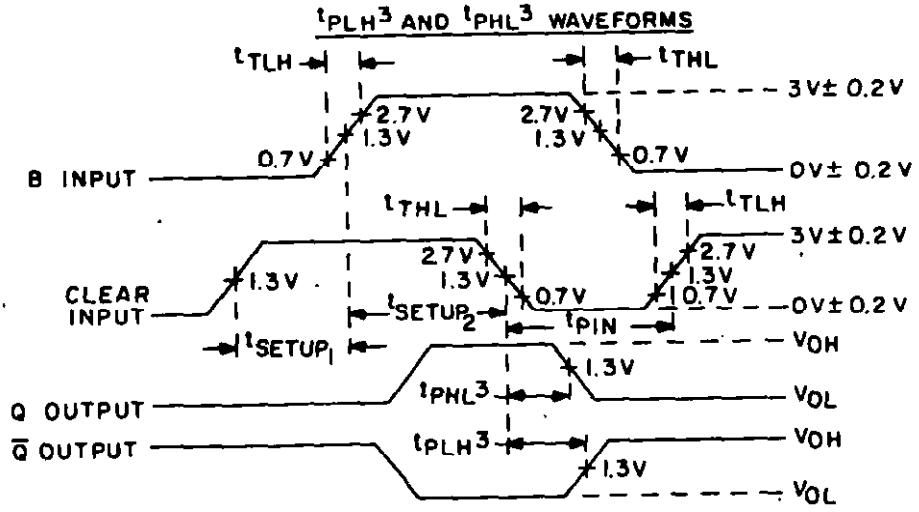


FIGURE 5. Switching test circuit and waveforms for device type 02.

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NOTES:
1. Pulse generator has the following characteristics: PRR ≤ 1.0 MHz, $t_{PIN} \geq 40$ ns; $t_{THL} \leq 6$ ns, $t_{TLH} \leq 15$ ns t_{SETUP1} (CLEAR INACTIVE) = 15 ns and $Z_{OUT} = 50\Omega$.
2. See table III notes for R_{ext}, C_{ext} values.
3. C_L = 50 pF ±10% including scope probe, wiring and stray capacitance, without package in test fixture.
4. All diodes are 1N3064 or 1N916 or equivalent.
5. Load circuit on a given output is only required where the specified test in table III indicates "OUT" on that output.
6. t_{SETUP} (max) shall be (≤ 50% of the typical output pulse width for the actual C_{ext} used) (see figure 2).

FIGURE 5. Switching test circuit and waveforms for device type 02 - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V; or low ≤0.7 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Case 2 / Test no.	1 / 2 / A1	2 / 3 / B1	3 / 4 / CLR1	4 / 5 / $\overline{Q1}$	5 / 7 / Q2	6 / 8 / Cext2	7 / 9 / Rext2	8 / 10 / GND	9 / 12 / A2	10 / 13 / B2	11 / 14 / CLR2	12 / 15 / $\overline{Q2}$	13 / 17 / Q1	14 / 18 / Cext1	15 / 19 / Rext1	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = 25°C	V_{OH}	3006	1	2.0 V	0.7 V	2.0 V	-.4 mA				GND					-.4 mA		GND	4.5 V	$\overline{Q1}$	2.5		V
	"	"	2	C	2.0 V	2.0 V					"								"	Q1	"		"
	"	"	3								"	2.0 V	0.7 V	2.0 V	-.4 mA				"	$\overline{Q2}$	"		"
	"	"	4					-.4 mA		GND	"	C	2.0 V	2.0 V					"	Q2	"		"
	V_{OL}	3007	5	2.0 V	0.7 V	2.0 V	4 mA				"					4 mA			"	Q1		0.4	"
	"	"	6	C	2.0 V	2.0 V					"							GND	"	$\overline{Q1}$		"	"
	"	"	7					4 mA			"	2.0 V	0.7 V	2.0 V					"	Q2		"	"
	"	"	8							GND	"	C	2.0 V	2.0 V	4 mA				"	$\overline{Q2}$		"	"
	V_{IC}		9	-18 mA							"								"	A1		-1.5	"
			10		-18 mA						"								"	B1		"	"
			11			-18 mA					"								"	CLR1		"	"
			12								"	-18 mA							"	A2		"	"
			13								"		-18 mA						"	B2		"	"
			14								"			-18 mA					"	CLR2		"	"
	I_{IL1} See note K	3009	15	0.4 V							"								5.5 V	A1	-160	-400	µA
	"	"	16		0.4 V						"								"	B1	"	"	"
	"	"	17			0.4 V					"								"	CLR1	"	"	"
	"	"	18								"	0.4 V							"	A2	"	"	"
	"	"	19								"		0.4 V						"	B2	"	"	"
	"	"	20								"			0.4 V					"	CLR2	"	"	"
	I_{IH1}	3010	21	2.7 V							"								"	A1		20	"
	"	"	22		2.7 V						"								"	B1		"	"
	"	"	23			2.7 V					"								"	CLR1		"	"
	"	"	24								"	2.7 V							"	A2		"	"
	"	"	25								"		2.7 V						"	B2		"	"
	"	"	26								"			2.7 V					"	CLR2		"	"
	I_{IH2}	"	27	5.5 V							"								"	A1		100	"
	"	"	28		5.5 V						"								"	B1		"	"
	"	"	29			5.5 V					"								"	CLR1		"	"
	"	"	30								"	5.5 V							"	A2		"	"
	"	"	31								"		5.5 V						"	B2		"	"
	"	"	32								"			5.5 V					"	CLR2		"	"
	I_{OS}	3011	33 CKT A, D	GND	GND	GND	GND				"								"	$\overline{Q1}$	-15	-100	mA
	"	"	33 CKT C	GND	GND	GND	GND				"								"		-30	-130	"
	"	"	34 CKT A, D	C	4.5 V	4.5 V					"					GND		GND	"	Q1	-15	-100	"
	"	"	34 CKT C	C	4.5 V	4.5 V					"					GND		GND	"		-30	-130	"
	"	"	35 CKT A, D								"	GND	GND	GND	GND				"	$\overline{Q2}$	-15	-100	"
	"	"	35 CKT C								"	GND	GND	GND	GND				"		-30	-130	"
	"	"	36 CKT A, D					GND		GND	"	C	4.5 V	4.5 V					"	Q2	-15	-100	"
	"	"	36 CKT C					GND		GND	"	C	4.5 V	4.5 V					"		-30	-130	"

See notes at end of this table.

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TABLE III. Group A inspection for device type 01 - Continued.

Terminal conditions (pins not designated may be high ≥2.0 V; or low ≤0.7 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Case 2 / Test no.	1 / 2 / A1	2 / 3 / B1	3 / 4 / CLR1	4 / 5 / $\overline{Q1}$	5 / 7 / Q2	6 / 8 / Cext2	7 / 9 / Rext2	8 / 10 / GND	9 / 12 / A2	10 / 13 / B2	11 / 14 / CLR2	12 / 15 / $\overline{Q2}$	13 / 17 / Q1	14 / 18 / Cext1	15 / 19 / Rext1	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = 25°C	I_{CC2}	3005 3005	37 38	5.5 V C	5.5 V 5.5 V	5.5 V 5.5 V					GND GND	5.5 V C	5.5 V 5.5 V	5.5 V 5.5 V					5.5 V 5.5 V	V_{CC} V_{CC}		20 20	mA mA
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = 125°C and V_{IC} tests omitted.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests omitted.																						
7 T_C = 25°C	Truth table tests	3014	39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55	A A B " " " " A A B B A " B	A B A " " B B A " " B B A " " B	B " " A B A A B B A A " " " " " "	H " L H " L H " " L H " " " " " "	L " H L " H L " " H L " " " " " "	F " " " " " " " " " " " " " " " "	F " " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " " "	A A B " " " " A A B B A " " B	A B A " " B B A " " B B A " " B	B " " A B A A B B A A " " " " " "	H " L H " L H " " L H " " " " " "	L " H L " H L " " H L " " " " " "	F " " " " " " " " " " " " " " " "	F " " " " " " " " " " " " " " " "	5.0 V " " " " " " " " " " " " " " " "	All outputs " " " " " " " " " " " " " " " "	See notes A, B, D, E and F		
8	Repeat subgroup 7 at T_C = 125°C and T_C = -55°C.																						
9 T_C = 25°C	t_{PLH1}	3003 (Fig. 5)	56 57	IN	5.0 V	5.0 V		OUT	F	F	GND "	IN	5.0 V	5.0 V		OUT	F	F	5.0 V "	A1 to Q1 A2 to Q2	5 "	38 38	ns "
	t_{PLH2}	" "	58 59	GND	IN	5.0 V		OUT	F	F	" "	GND	IN	5.0 V		OUT	F	F	" "	B2 to Q2 B1 to Q1	" "	49 49	" "
	t_{PLH3}	" "	60 61	GND	IN	IN	OUT		L	L	" "	GND	IN	IN	OUT		L	L	" "	CLR1 to $\overline{Q1}$ CLR2 to $\overline{Q2}$	" "	50 "	" "
	t_{PHL1}	" "	62 63	IN	5.0 V	5.0 V	OUT		F	F	" "	IN	5.0 V	5.0 V	OUT		F	F	" "	A2 to $\overline{Q2}$ A1 to $\overline{Q1}$	" "	" "	" "
	t_{PHL2}	" "	64 65	GND	IN	5.0 V	OUT		F	F	" "	GND	IN	5.0 V	OUT		F	F	" "	B1 to $\overline{Q1}$ B2 to $\overline{Q2}$	" "	61 61	" "
	t_{PHL3}	" "	66 67	GND	IN	IN		OUT	L	L	" "	GND	IN	IN		OUT	L	L	" "	CLR2 to $\overline{Q2}$ CLR1 to $\overline{Q1}$	" "	32 32	" "
	$t_{p(min)}$	" " " "	68 69 70 71	IN GND	5.0 V IN	5.0 V 5.0 V		OUT OUT	OPEN OPEN	G G	" " " "	IN GND	5.0 V IN	5.0 V 5.0 V		OUT OUT	Open Open	G G	" " " "	Q1 Q1 Q2 Q2	- - -	205 " " "	" " " "

See notes at end of this table.

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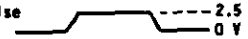
TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; or low ≤0.7 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	A1	B1	CLR1	$\overline{Q1}$	Q2	Cext2	Rext2	GND	A2	B2	CLR2	$\overline{Q2}$	Q1	Cext1	Rext1	V_{CC}				
9 T_C = 25°C	t_{p4}	3003 (Fig. 5)	72					OUT	I	I	GND	GND	IN	5.0 V					5.0 V	Q2	3.5	6.0	µs
		"	73					OUT	I	I	"	IN	5.0 V	5.0 V					"	Q2	"	"	"
		"	74	IN	5.0 V	5.0 V					"					OUT	I	I	"	Q1	"	"	"
		"	75	GND	IN	5.0 V					"					OUT	I	I	"	Q1	"	"	"
10 T_C = 125°C	Same tests and terminal conditions as subgroup 9, except T_C = 125°C and limits are as follows: t_{PLH1} is 5 to 57 ns; t_{PLH2} is 5 to 74 ns; t_{PLH3} is 5 to 75 ns; t_{PHL1} is 5 to 75 ns; t_{PHL2} is 5 to 92; t_{PHL3} is 5 to 48 ns; $t_{p(MIN)}$ is 308 ns; and t_{p4} is 3.0 to 6.25 µs.																						
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

NOTES:

A. V_{IN} = 3.0 V minimum.

B. V_{IN} = 0.0 V or GND.

C. Apply input pulse  2.5 V min/5.5 V max. 0 V

D. Test numbers 39 through 55 shall be run in sequence.

E. Output voltages shall be either:

H > 1.5 V; L < 1.5 V

F. R_{ext} = 5 kΩ minimum to 180 kΩ maximum, connected to V_{CC}; C_{ext} ≤1,000 µF, connected to R_{ext} terminal.

G. R_{ext} = 5 kΩ ±10%, connected to V_{CC}.

I. R_{ext} = 10 kΩ ±10%, connected to V_{CC}; C_{ext} = 1,000 pF ±10% connected to R_{ext} terminal.

J. During subgroups 9, 10, and 11 testing, R_{ext} and C_{ext} may remain applied on the side of the device not under test if desired.

K. For circuit D, I_{IL1} limits are 120 mA to 360 mA.

L. R_{ext} = 10 kΩ ±10% connected to V_{CC}; C_{ext} ≥45 pF connected to R_{ext} terminal.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥2.0 V; or low ≤0.7 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Case 2 / Test no.	1 / 2 / A1	2 / 3 / B1	3 / 4 / CLR1	4 / 5 / $\overline{Q1}$	5 / 7 / Q2	6 / 8 / Cext2 See L	7 / 9 / Rext2 See L	8 / 10 / GND	9 / 12 / A2	10 / 13 / B2	11 / 14 / CLR2	12 / 15 / $\overline{Q2}$	13 / 17 / Q1	14 / 18 / Cext1 See L	15 / 19 / Rext1 See L	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = 25°C	V_{OH}	3006	1	2.0 V	0.7 V	2.0 V	-.4 mA				GND								4.5 V	$\overline{Q1}$	2.5		V
	"	"	2	C	2.0 V	2.0 V					"								"	Q1	"		"
	"	"	3								"	2.0 V	0.7 V	2.0 V	-.4 mA	-.4 mA	GND	GND	"	$\overline{Q2}$	"		"
	"	"	4					-.4 mA	GND	GND	"	C	2.0 V	2.0 V					"	Q2	"		"
	V_{OL}	3007	5	2.0 V	0.7 V	2.0 V	4 mA				"					4 mA			"	Q1		0.4	"
	"	"	6	C	2.0 V	2.0 V					"								"	$\overline{Q1}$		"	"
	"	"	7					4 mA			"	2.0 V	0.7 V	2.0 V			GND	GND	"	Q2		"	"
	"	"	8						GND	GND	"	C	2.0 V	2.0 V	4 mA				"	$\overline{Q2}$		"	"
	V_{IC}		9	-18 mA							"								"	A1		-1.5	"
			10		-18 mA						"								"	B1		"	"
			11			-18 mA					"								"	CLR1		"	"
			12								"	-18 mA							"	A2		"	"
			13								"		-18 mA						"	B2		"	"
			14								"			-18 mA					"	CLR2		"	"
	I_{IL2}	3009	15 CKT A, C, D	0.4 V	4.5 V	4.5 V					"								5.5 V	A1	-100	-340	uA
	"	"	15 CKT B	"	"	"					"								"	A1	-30	-300	"
	"	"	15 CKT E	"	"	"					"								"	A1	-160	-400	"
	"	"	16 CKT A, C	GND	0.4 V	"					"								"	B1	-160	-580	"
	"	"	16 CKT B	"	"	"					"								"	B1	-30	-300	"
	"	"	16 CKT E, D	"	"	"					"								"	B1	-200	-440	"
	"	"	17 CKT A,C,D,E	"	4.5 V	GND					"						GND	GND	"	CLR1	-200	-680	"
	"	"	17 CKT B	"	4.5 V	GND					"						GND	GND	"	CLR1	-30	-300	"
	"	"	18 CKT A, C, D								"	0.4 V	4.5 V	4.5 V					"	A2	-100	-340	"
	"	"	18 CKT B								"	"	"	"					"	A2	-30	-300	"
	"	"	18 CKT E								"	"	"	"					"	A2	-160	-400	"
	"	"	19 CKT A, C								"	GND	0.4 V	"					"	B2	-160	-580	"
	"	"	19 CKT B								"	"	"	"					"	B2	-30	-300	"
	"	"	19 CKT E, D								"	"	"	"					"	B2	-200	-440	"
	"	"	20 CKT A,C,D,E						GND	GND	"	"	4.5 V	GND					"	CLR2	-200	-680	"
	"	"	20 CKT B						GND	GND	"	"	4.5 V	GND					"	CLR2	-30	-300	"
	I_{IH1}	3010	21	2.7 V							"								"	A1		20	"
	"	"	22	4.5 V	2.7 V	GND					"								"	B1		"	"
	"	"	23	4.5 V	GND	2.7 V					"								"	CLR1		"	"
	"	"	24								"	2.7 V							"	A2		"	"
	"	"	25								"	4.5 V	2.7 V	GND					"	B2		"	"
	"	"	26								"	4.5 V	GND	2.7 V					"	CLR2		"	"

See notes at end of this table.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; or low ≤0.7 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Case 2 / Test no.	1 / 2 / A1	2 / 3 / B1	3 / 4 / CLR1	4 / 5 / $\overline{Q1}$	5 / 7 / Q2	6 / 8 / Cext2 See L	7 / 9 / Rext2 See L	8 / 10 / GND	9 / 12 / A2	10 / 13 / B2	11 / 14 / CLR2	12 / 15 / $\overline{Q2}$	13 / 17 / Q1	14 / 18 / Cext1 See L	15 / 19 / Rext1 See L	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = 25°C	I_{IH2}	3010	27	5.5 V							GND								5.5 V	A1		100	µA
	"	"	28	4.5 V	5.5 V	GND					"								"	B1		"	"
	"	"	29	4.5 V	GND	5.5 V					"								"	CLR1		"	"
	"	"	30								"	5.5 V							"	A2		"	"
	"	"	31								"	4.5 V	5.5 V	GND					"	B2		"	"
	"	"	32								"	4.5 V	GND	5.5 V					"	CLR2		"	"
	I_{OS}	3011	33 CKT A,B,D,E	C	4.5 V	4.5 V					"					GND	GND	GND	"	Q1	-15	-100	mA
	"	"	33 CKT C	C	4.5 V	4.5 V					"					GND	GND	GND	"	Q1	-30	-130	"
	"	"	34 CKT A,B,D,E								"	GND	GND	GND	GND					$\overline{Q2}$	-15	-100	"
	"	"	34 CKT C								"	GND	GND	GND	GND				"	$\overline{Q2}$	-30	-130	"
	"	"	35 CKT A,B,D,E					GND	GND	GND	"	C	4.5 V	4.5 V					"	Q2	-15	-100	"
	"	"	35 CKT C					GND	GND	GND	"	C	4.5 V	4.5 V					"	Q2	-30	-130	"
	"	"	36 CKT A,B,D,E	GND	GND	GND	GND				"								"	$\overline{Q1}$	-15	-100	"
	"	"	36 CKT C	GND	GND	GND	GND				"								"	$\overline{Q1}$	-30	-130	"
	I_{CC1}	3005	37	5.5 V	GND	5.5 V					"	5.5 V	GND	5.5 V					"	V_{CC}		11.0	"
	I_{CC3}	3005	38	C	5.5 V	5.5 V					"	C	5.5 V	5.5 V					"	V_{CC}		27.0	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = 25°C	Truth table tests	3014	39	A	A	B	H	L	F	F	GND	A	A	B	H	L	F	F	5.0 V	All outputs	See notes A, B, D, E, and F		
	"	"	40	A	B	"	"	"	"	"	"	A	B	"	"	"	"	"	"	"	"	"	
	"	"	41	B	A	"	"	"	"	"	"	B	A	"	"	"	"	"	"	"	"	"	
	"	"	42	"	"	A	L	H	"	"	"	"	"	A	L	H	"	"	"	"	"	"	
	"	"	43	"	"	B	H	L	"	"	"	"	"	B	H	L	"	"	"	"	"	"	
	"	"	44	"	B	B	"	"	"	"	"	"	B	B	"	"	"	"	"	"	"	"	
	"	"	45	"	B	A	"	"	"	"	"	"	B	A	"	"	"	"	"	"	"	"	
	"	"	46	"	A	A	L	H	"	"	"	"	A	A	L	H	"	"	"	"	"	"	
	"	"	47	"	"	B	H	L	"	"	"	"	"	B	H	L	"	"	"	"	"	"	
	"	"	48	A	"	B	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"	"	
	"	"	49	A	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	
	"	"	50	B	"	A	L	H	"	"	"	"	"	"	"	H	"	"	"	"	"	"	
	"	"	51	"	"	B	H	L	"	"	"	"	"	"	"	L	"	"	"	"	"	"	
	"	"	52	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	
	"	"	53	"	"	"	"	"	"	"	"	B	A	A	L	"	"	"	"	"	"	"	
	"	"	54	"	B	"	"	"	"	"	"	B	B	B	H	"	"	"	"	"	"	"	
	"	"	55	A	B	A	"	"	"	"	"	A	B	A	"	"	"	"	"	"	"	"	
	"	"	56	"	A	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"	
	"	"	57	"	B	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	
	"	"	58	B	B	"	"	"	"	"	"	B	B	"	"	"	"	"	"	"	"	"	

See notes at end of this table.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; or low ≤0.7 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Case 2 / Test no.	1 / 2 / A1	2 / 3 / B1	3 / 4 / CLR1	4 / 5 / $\overline{Q1}$	5 / 7 / Q2	6 / 8 / Cext2	7 / 9 / Rext2	8 / 10 / GND	9 / 12 / A2	10 / 13 / B2	11 / 14 / CLR2	12 / 15 / $\overline{Q2}$	13 / 17 / Q1	14 / 18 / Cext1	15 / 19 / Rext1	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
8	Repeat subgroup 7 at T_C = 125°C and -55°C.																						
9 T_C = 25°C	t_{PLH1}	3003 (Fig. 6)	59 60	IN	5.0 V	5.0 V		OUT	F	F	GND "	IN	5.0 V	5.0 V		OUT	F	F	5.0 V "	A1 to Q1 A2 to Q2	5 "	75 75	ns "
	t_{PLH2}	" "	61 62	GND	IN	5.0 V		OUT	F	F	" "	GND	IN	5.0 V		OUT	F	F	" "	B2 to Q2 B1 to Q1	" "	60 60	" "
	t_{PLH3}	" "	63 64	GND	IN	IN	OUT		N	N	" "	GND	IN	IN	OUT		N	N	" "	CLR1 to $\overline{Q1}$ CLR2 to $\overline{Q2}$	" "	70 70	" "
	t_{PHL1}	" "	65 66	IN	5.0 V	5.0 V	OUT		F	F	" "	IN	5.0 V	5.0 V	OUT		F	F	" "	A2 to $\overline{Q2}$ A1 to $\overline{Q1}$	" "	85 85	" "
	t_{PHL2}	" "	67 68	GND	IN	5.0 V	OUT		F	F	" "	GND	IN	5.0 V	OUT		F	F	" "	B1 to $\overline{Q1}$ B2 to $\overline{Q2}$	" "	70 70	" "
	t_{PHL3}	" "	69 70	GND	IN	IN		OUT	N	N	" "	GND	IN	IN		OUT	N	N	" "	CLR2 to $\overline{Q2}$ CLR1 to $\overline{Q1}$	" "	60 60	" "
	t_{p1}	" " " " " " " "	71 72 73 74 75 76 77 78	IN IN GND GND	5.0 V 5.0 V IN IN	5.0 V	 OUT OUT 	 OUT OUT	 OPEN " " "	 K " " "	" " " " " " " "	 IN IN GND "	 5.0 V 5.0 V IN "	 5.0 V " " "	 OUT OUT	OUT OUT	OPEN " " 	K " " 	" " " " " " " "	Q1 $\overline{Q1}$ $\overline{Q1}$ Q1 Q2 $\overline{Q2}$ $\overline{Q2}$ Q2	20 " " " " " " "	75 " " " " " " "	" " " " " " " "
	t_{p2}	" " " " " " " "	79 80 81 82 83 84 85 86	 IN IN GND "	 5.0 V 5.0 V IN "	 5.0 V " " "	 OUT OUT 	OUT OUT 	G " " " 	G " " " 	" " " " " " " "	" " IN IN 	" " 5.0 V 5.0 V 	" " " " 	 OUT OUT 	 OUT OUT	 G " " 	 G " " 	" " " " " " " "	Q2 $\overline{Q2}$ $\overline{Q2}$ Q2 Q1 $\overline{Q1}$ $\overline{Q1}$ Q1	70 " " " " " " "	160 " " " " " " "	" " " " " " " "
	t_{p3}	" " " " " " " "	87 88 89 90 91 92 93 94	" " IN IN	" " 5.0 V 5.0 V	" " " "	 OUT OUT 	 OUT OUT	 I " " "	 I " " "	" " " " " " " "	 IN IN GND "	 5.0 V 5.0 V IN "	 5.0 V " " "	 OUT OUT	OUT OUT	I " " 	I " " 	" " " " " " " "	Q1 $\overline{Q1}$ $\overline{Q1}$ Q1 Q2 $\overline{Q2}$ $\overline{Q2}$ Q2	600 " " " " " " "	775 " " " " " " "	" " " " " " " "
	t_{p5}	" " "	95 96 97					OUT	J " "	J " "	" " "	" " IN	" " 5.0 V	" " "	 OUT OUT				" " "	Q2 $\overline{Q2}$ $\overline{Q2}$	5.7 " "	8.0 " "	ms " "

See notes at end of this table.

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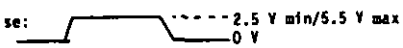
TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; or low ≤0.7 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	A1	B1	CLR1	$\overline{Q1}$	Q2	Cext2	Rext2	GND	A2	B2	CLR2	$\overline{Q2}$	Q1	Cext1	Rext1	V_{CC}				
9 T_C = 25°C	t_{p5}	3003 (Fig. 6)	98					OUT				IN	5.0 V	5.0 V					5.0 V	Q2	5.7	8.0	ms
		"	99	IN	5.0 V	5.0 V										OUT	J	J	"	Q1	"	"	"
		"	100	IN	5.0 V	"	OUT										"	"	"	$\overline{Q1}$	"	"	"
		"	101	GND	IN	"	OUT										"	"	"	$\overline{Q1}$	"	"	"
		"	102	GND	IN	"										OUT	"	"	"	Q1	"	"	"
10 T_C = 125°C	Same tests and terminal conditions as subgroup 9, except T_C = 125°C and limits are as follows: t_{PLH1} is 5 to 113 ns; t_{PLH2} is 5 to 90 ns; t_{PLH3} is 5 to 105 ns; t_{PHL1} is 5 to 128 ns; t_{PHL2} is 5 to 105 ns; t_{PHL3} is 5 to 90 ns; t_{p1} is 20 to 91 ns; t_{p2} is 70 to 195 ns; t_{p3} is 600 to 850 ns; and t_{p5} is 5.5 to 8.5 ms.																						
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

NOTES:

A. V_{IN} = 3.0 V minimum.

B. V_{IN} = 0.0 V or GND.

C. Apply input pulse: 

D. Test numbers 39 through 58 shall be run in sequence.

E. Output voltages shall be either:

H > 1.5 V; L < 1.5 V

F. R_{ext} = 1.4 kΩ minimum to 70 kΩ maximum, connected to V_{CC}; C_{ext} ≤1,000 µF, connected to R_{ext} terminal.

G. R_{ext} = 2 kΩ ±10%, connected to V_{CC}; C_{ext} = 80 pF ±10%, connected to R_{ext} terminal.

I. R_{ext} = 10 kΩ ±10%, connected to V_{CC}; C_{ext} = 100 pF ±10%, connected to R_{ext} terminal.

J. R_{ext} = 10 kΩ ±10%, connected to V_{CC}; C_{ext} = 1.0 µF ±10%, connected to R_{ext} terminal.

K. R_{ext} = 2 kΩ ±10%, connected to V_{CC}.

L. Note F may apply during subgroups 1, 2, and 3 testing if desired.

M. During subgroups 9, 10, and 11 testing, R_{ext} and C_{ext} may remain applied on the side of the device not under test if desired.

N. R_{ext} = 10 kΩ ±10% connected to V_{CC}; C_{ext} ≥45 pF connected to R_{ext} terminal.

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D / Case 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits Min	Limits Max	Unit
				2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	A1	A2	B1	B2	CLR	$\overline{Q}$	GND	Q	Rint	NC	Cext (see I)	NC	Rext $\overline{Cext}$ (see I)	V_{CC}				
1 T_C = 25°C	V_{OH}	3006 3006	1 2	0.7 V		2.0 V	J 0.7 V	2.0 V "	-.4 mA	GND "	-.4 mA	GND 4.5 V				GND	4.5 V "	Q $\overline{Q}$	2.5 2.5		V "
	V_{OL}	3007 "	3 4	0.7 V		2.0 V	0.7 V J	" "	4 mA	" "	4 mA	4.5 V GND				GND	" "	Q $\overline{Q}$		0.4 0.4	" "
	V_{IC}	" " " " "	5 6 7 8 9	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA		" " " " "							" " " " "	A1 A2 B1 B2 CLR		-1.5 " " " "	" " " " "
	I_{IL1} See note C	3009 " " " "	10 11 12 13 14	0.4 V 4.5 V	4.5 V 0.4 V	0.4 V 4.5 V	4.5 V 0.4 V	0.4 V		" " " " "							5.5 V " " " "	A1 A2 B1 B2 CLR	-160 " " " "	-400 " " " "	µA " " " "
	I_{IH1}	3010 " " " "	15 16 17 18 19	2.7 V GND	GND 2.7 V	2.7 V GND	GND 2.7 V	2.7 V		" " " " "							" " " " "	A1 A2 B1 B2 CLR		20 " " " "	" " " " "
	I_{IH2}	" " " " "	20 21 22 23 24	5.5 V GND	GND 5.5 V	5.5 V GND	GND 5.5 V	5.5 V		" " " " "							" " " " "	A1 A2 B1 B2 CLR		100 " " " "	" " " " "
	I_{OS}	3011 " " "	25 CKT A,D 25 CKT C 26 CKT A,D 26 CKT C	GND GND		4.5 V 4.5 V	J J GND GND	4.5 V " " "	 GND GND	" " " "	GND GND	GND GND 4.5 V 4.5 V				GND GND	" " " "	Q Q $\overline{Q}$ $\overline{Q}$	-15 -30 -15 -30	-100 -130 -100 -130	mA " " "
	I_{CC1}	3005	27	5.5 V	5.5 V	GND	GND	5.5 V		"		5.5 V					"	V_{CC}		11	"
	I_{CC2}	3005	28	J	J	5.5 V	5.5 V	5.5 V		"		5.5 V					"	V_{CC}		11	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				

See notes at end of this table.

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TABLE III. Group A (inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	A1	A2	B1	B2	CLR	$\overline{Q}$	GND	Q	Rint	NC	Cext	NC	Rext/Cext	V_{CC}				
7 T_C = 25°C	Truth table test	3014	29	B	B	B	B	B	H	GND	L			K		K	5.0 V	All outputs		See notes A, B	
		"	29A	A	A	B	B	A	L	"	H			"		"	"	"			
		"	30	A	A	A	A	A	H	"	L			"		"	"	"			
		"	31	B	B	A	A	B	L	"	H			"		"	"	"			
		"	32	"	"	"	"	A	H	"	L			"		"	"	"			
		"	33	"	A	B	B	B	L	"	H			"		"	"	"			
		"	34	"	"	B	A	A	H	"	L			"		"	"	"			
		"	35	"	"	A	B	B	L	"	H			"		"	"	"			
		"	36	"	"	"	B	A	H	"	L			"		"	"	"			
		"	37	"	"	"	A	B	L	"	H			"		"	"	"			
		"	38	"	"	"	A	A	H	"	L			"		"	"	"			
		"	39	"	"	"	"	B	L	"	H			"		"	"	"			
		"	40	A	"	"	"	A	H	"	L			"		"	"	"			
		"	41	A	"	"	"	B	L	"	H			"		"	"	"			
		"	42	B	"	"	"	A	H	"	L			"		"	"	"			
		"	43	A	"	"	"	B	L	"	H			"		"	"	"			
		"	44	"	"	"	"	A	H	"	L			"		"	"	"			
		"	45	"	B	"	"	A	L	"	H			"		"	"	"			
8	Repeat subgroup 7 at T_C = 125°C and T_C = -55°C.																				
9 T_C = 25°C	t_{PHL1}	3003 (Fig. 5)	46	IN	5.0 V	5.0 V	5.0 V	5.0 V	OUT	GND				K		K	5.0 V	A1 to $\overline{Q}$	5	50	ns
	t_{PHL2}	"	47	GND	GND	IN	5.0 V	5.0 V	OUT	"				K		K	"	B1 to $\overline{Q}$		61	"
	t_{PHL3}	"	48	GND	GND	5.0 V	IN	IN		"	OUT			M		M	"	CLR to Q		32	"
	t_{PLH1}	"	49	IN	5.0 V	5.0 V	5.0 V	5.0 V		"	OUT			K		K	"	A1 to Q		38	"
	t_{PLH2}	"	50	GND	GND	IN	5.0 V	5.0 V		"	OUT			K		K	"	B1 to Q		49	"
	t_{PLH3}	"	51	GND	GND	5.0 V	IN	IN	OUT	"				M		M	"	CLR to $\overline{Q}$		50	"
	$t_{p(MIN)}$	"	52	IN	5.0 V	5.0 V	5.0 V	5.0 V		"	OUT					G	"	Q		205	"
	t_{p4}	"	53	GND	GND	IN	5.0 V			"	OUT			F		N	"	Q	3.5	6.0	µs
10 T_C = 125°C	Same tests and terminal conditions as subgroup 9, except T_C = 125°C and limits are as follows: t_{PHL1} is 50 to 75 ns; t_{PHL2} is 5 to 92 ns; t_{PHL3} is 5 to 48 ns; t_{PLH1} is 5 to 57 ns; t_{PLH2} is 5 to 74 ns; t_{PLH3} is 5 to 75 ns; $t_{p(MIN)}$ is 308 ns; and t_{p4} is 3.0 to 6.25 µs.																				
11	Same tests, terminal conditions, and limits as for subgroup 10 except T_C = -55°C.																				

NOTES:

A. V_{IN} = 3.0 V minimum.
B. V_{IN} = 0.0 V or GND.
C. For circuit D, I_{IL1} limits are 120 to 360 mA.
D. Test numbers 29 through 45 shall be run in sequence.
E. Output voltages shall be either:

H > 1.5 V; L < 1.5 V

F. C_{ext} connected to R_{ext}/C_{ext} through a 1,000 pF ±10% capacitor.

G. R_{ext}/C_{ext} connected to V_{CC} through a 5 kΩ ±10% resistor.
I. Note K may apply during subgroups 1, 2, and 3 testing if desired.

J. Apply input pulse:  2.5 V minimum to 5.5 V maximum, 0 V

K. R_{ext}/C_{ext} connected to V_{CC} through a 5 kΩ to 180 kΩ resistor, and C_{ext} connected to R_{ext}/C_{ext} through a ≤1,000 µF capacitor.
N. R_{ext}/C_{ext} connected to V_{CC} through a 10 kΩ ±10% resistor.
M. C_{ext} connected to R_{ext}/C_{ext} through A ≥45 pF capacitor. R_{ext}/C_{ext} connected to V_{CC} through A 10 kΩ ±10% resistor.

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4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
I_{IN}	Current flowing into an input terminal.
V_{IC}	Input clamp voltage.
V_{IN}	Voltage level at an input terminal.

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

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Military device type	Generic-industry type
01	54LS123, 74LS123
02	54LS221, 74LS221
03	54LS122, 74LS122

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.8 Manufacturers' designators. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuit					
	Commercial Type	A Texas Instruments, Incorporated	B Signetics Corporation	C National Semi-conductor Corpo-ration	D Motorola Incorpo-rated	E Raytheon Co.
01	54LS123	X		X	X	
02	54LS221	X	X	X	X	
03	54LS122	X		X	X	X

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-0633)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/314B	Microcircuits, Digital, Low Power Schottky, TTL, Monostable Multivibrators, Monolithic Silicon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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